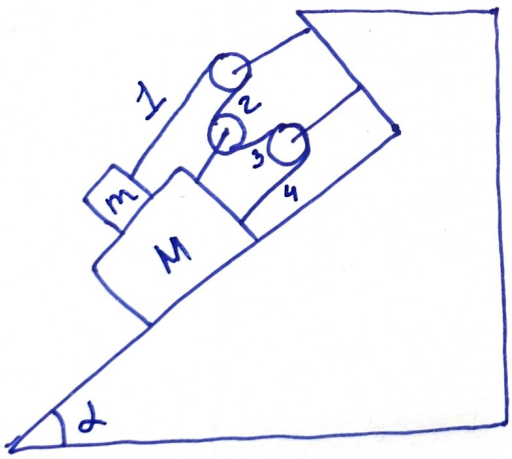
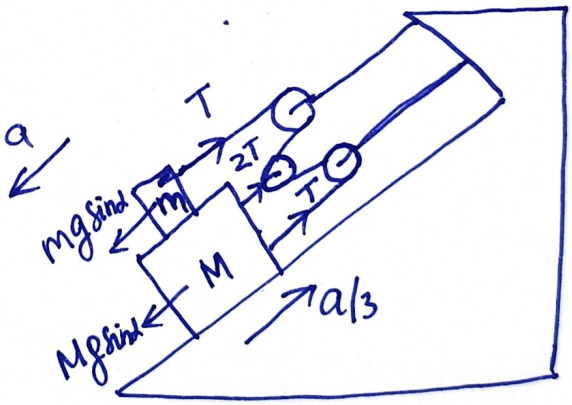
1- masala Yechimi

ⓐ Agar m yuk pastga harakatlansa 1- uchastka uzayadi. 2,3,4 qismlar esa bir xilda qisqaradi. Bunga ko'ra M yuk ko'tariladi. Aksincha m yuk yuqoriga harakatlansa 1-qism qisqaradi. 2,3,4 qismlar esa uzayadi „Bir xil uzunlikka". M yuk esa pastga harakatlanadi. Ko'rinib turibdiki ikkala holatda ham ular o'zaro qarama-qarshi yo'nalishda harakatlanmoqda ularning qanaqa yo'nalishda harakatlanishi massalar nisbatiga ham bog'liq. Qaysidir ~~m~~ M yuk massasining kattaligi (aniqrog'i cheksiz kattaligi $m \gg M$ yoki $M \gg m$) harakat yo'nalishini belgilaydi. (1 ball) (1 ball)

ⓑ Umumiy gorizontal uchastkalar 1,2,3,4 yig'indisi o'zgarmaydi. Agar 1 qism 3l masofaga o'zgarsa 2,3,4 uchastkalar bir xil l masofaga siljiydi. Bundan m va M larning ko'chishlar nisbati 3:1. (1 ball) Xuddi shunga mutanosib tarzda $\frac{v_m}{v_M} = 3$ (1 ball)



1- rasm



2- rasm

ⓒ Agar m brusok qimirlamasa M brusok ham siljimaydi. Bu degani unga ta'sir qiluvchi kuchlarning vektorlar yig'indisi nolga teng. 2- rasmga ko'ra $Mg\sin\alpha = 3T$ (1 ball)

$$T = \frac{Mg\sin\alpha}{3} = \frac{2 \cdot 10 \cdot \frac{1}{2}}{3} \approx 3{,}3\ H$$ (1 ball)

Chizma uchun (1 ball) (Chizmada teng lanish bo'lishi shart emas)

d) Agar m massali yuk a tezlanish bilan yuqoriga ketsa (yoki pastga) M massali yuk a/3 tezlanish bilan pastga tushadi (yuqoriga). $\frac{v_m}{v_M}=3$ dan buni yaqqol ko'rish mumkin. Endi ikkala jism uchun Nyutonning 2-qonunini yozamiz.

$ma = mg\sin\alpha - T$ (1 ball)

$M\frac{a}{3} = 3T - Mg\sin\alpha$ (1 ball)

Sistemani yechish orqali ani topib olamiz.

$$a = \frac{|gm-3M|}{gm+M}\cdot g\cdot\sin\alpha \approx \frac{1}{4}\frac{m}{c^2}$$ (1 ball)

$M>3m$ shartda jismlar sistemasi 2-rasmdagidan teskari yo'nalishda harakatlanishni boshlaydi